

PRAKTIKUM BASIS DATA

MODUL 9

Scripting Procedure dan Function



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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Jurnal

1. Buat tabel employees dari hremployees

Query

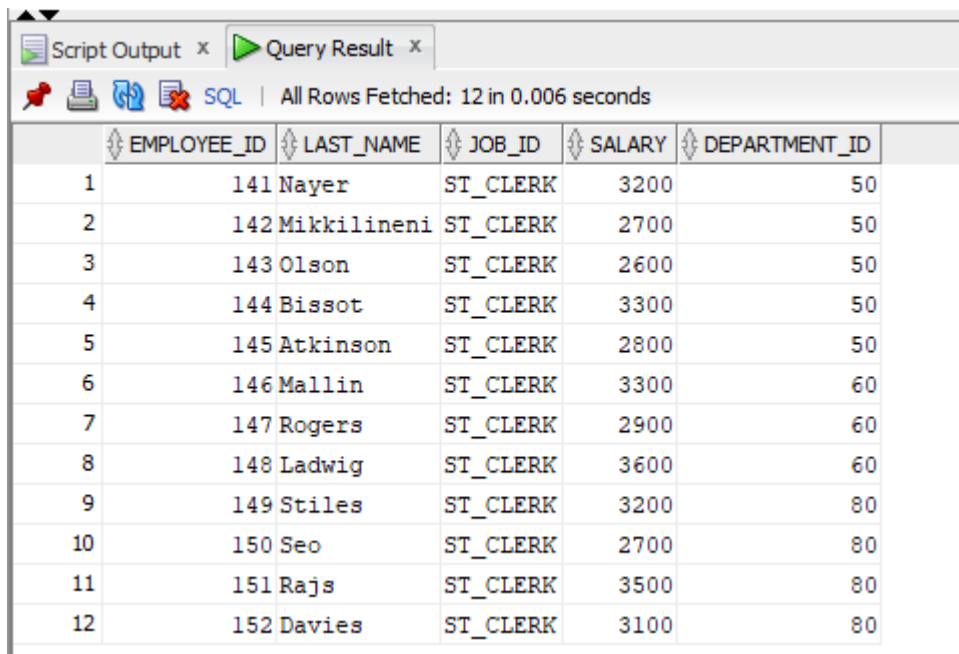
```
-- 1. Buat tabel employees dari hremployees
CREATE TABLE employees AS SELECT * FROM hremployees;
```

2. Select (tampilkan) semua data employees

Query

```
-- 2. Select semua data employees
SELECT * FROM employees;
```

Output



	EMPLOYEE_ID	LAST_NAME	JOB_ID	SALARY	DEPARTMENT_ID
1	141	Nayer	ST_CLERK	3200	50
2	142	Mikkilineni	ST_CLERK	2700	50
3	143	Olson	ST_CLERK	2600	50
4	144	Bissot	ST_CLERK	3300	50
5	145	Atkinson	ST_CLERK	2800	50
6	146	Mallin	ST_CLERK	3300	60
7	147	Rogers	ST_CLERK	2900	60
8	148	Ladwig	ST_CLERK	3600	60
9	149	Stiles	ST_CLERK	3200	80
10	150	Seo	ST_CLERK	2700	80
11	151	Rajs	ST_CLERK	3500	80
12	152	Davies	ST_CLERK	3100	80

3. Buat procedure increase_salary

Query

```
-- 3. Buat procedure increase_salary
CREATE OR REPLACE PROCEDURE increase_salary (
    department_id_in IN employees.department_id%TYPE,
    increase_pct_in IN NUMBER)
IS
BEGIN
    FOR employee_rec
    IN (SELECT employee_id
        FROM employees
        WHERE department_id = increase_salary.department_id_in)
    LOOP
        UPDATE employees emp
        SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in
        WHERE emp.employee_id = employee_rec.employee_id;
        DBMS_OUTPUT.PUT_LINE('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;
```

Output

```
-- 3. Buat procedure increase_salary
CREATE OR REPLACE PROCEDURE increase_salary (
    department_id_in IN employees.department_id%TYPE,
    increase_pct_in IN NUMBER)
IS
BEGIN
    FOR employee_rec
    IN (SELECT employee_id
        FROM employees
        WHERE department_id = increase_salary.department_id_in)
    LOOP
        UPDATE employees emp
        SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in
        WHERE emp.employee_id = employee_rec.employee_id;
        DBMS_OUTPUT.PUT_LINE('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;
/
```

4. Eksekusi procedure + ROLLBACK

Query

```
-- 4. Eksekusi procedure + ROLLBACK
BEGIN
    increase_salary(90, .50);
    ROLLBACK;
END;
/

SELECT * FROM employees WHERE department_id = 90;
```

Output

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90

5. Apa yang terjadi jika statement ROLLBACK dihapus pada sql code no. 4?
Mengembalikan nilai ke awal (jika menggunakan statement). Tapi jika tidak menggunakan statement rollback, ia akan mengkalkulasikan datanya.